

Protecting WiFi Human Sensing Against Eavesdropping via RIS-enabled Physical Mask

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Abstract—WiFi signals inherently carry rich information about their surroundings, which inevitably provides eavesdroppers with opportunities to infer private information. Scrambling these signals has proven as an effective method to thwart eavesdropping, but existing approaches achieving signal scrambling either require modifying standard WiFi hardware or degrade legitimate sensing performance. In this paper, we propose RISEC, a hardware–software co-design that leverages a reconfigurable intelligent surface (RIS) to protect WiFi-based human sensing. At its core, RISEC fully exploits the RIS’s channel programmability to apply random phase modulation to scramble WiFi signals and distort the corresponding sensing information. Such random modulation functions as a dynamic physical mask in the channel. We delicately devise the mask in time and spatial domains to ensure sufficient signal scrambling. Besides, we also encode essential parameters in the mask to help legitimate users uncover the mask and reverse the scrambling. Further, by using our dedicated algorithm, legitimate users can accurately recover sensing information from the superposition of RIS-initiated and Tx-initiated signals. We implement RISEC and evaluate it under various settings. Results show that RISEC significantly reduces an eavesdropper’s sensing accuracy to around random-guess levels in password inference (from 94.82% down to 0.25%) and activity recognition (from 92.38% down to 19.36%), respectively.

Index Terms—Secure WiFi sensing, signal scrambling, sensing information protection

I. INTRODUCTION

With the rapid development of WiFi sensing techniques and the ubiquity of existing WiFi infrastructure, people can conveniently enjoy diverse sensing applications [1]–[4]. For instance, one could use a handy WiFi device to extract and analyze channel state information (CSI) from received signals to support indoor localization [5], [6], health monitoring [7], [8], and many other practical sensing services.

Despite the advances, the inherently open nature of WiFi signals provides eavesdroppers opportunities to infer private information. As shown in Fig. 1, an eavesdropper can deploy a WiFi receiver (Rx) to record signals and analyze CSI to infer a victim’s sensitive information, such as passwords [9], [10] and daily activity [11]. The exposure of such information may lead to financial or medical fraud, and thus bring economic

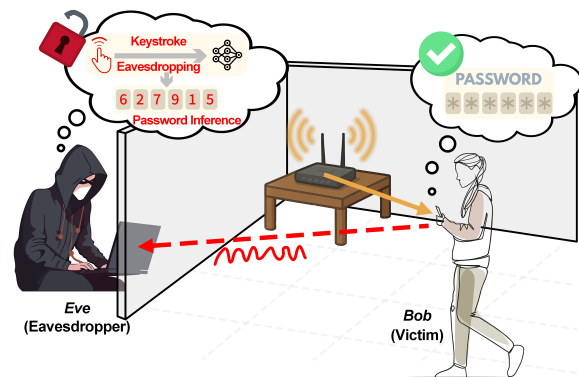


Fig. 1. An example of an eavesdropper inferring a victim’s password by analyzing received WiFi signals.

or physical harm to the victim. What’s worse, these eavesdroppers usually operate passively and generate no observable traffic, so it is difficult to detect them for defense [11], [12].

To thwart such eavesdropping, existing works adapt to scramble WiFi signals by deliberately altering signal characteristics. By doing so, the private information carried by WiFi signals will be distorted, and the CSI extracted by eavesdroppers will be meaningless. In summary, existing methods that achieve signal scrambling can be categorized as follows:

One type of method scrambles WiFi signals based on WiFi devices. Specifically, some of them adjust MIMO configurations or alter the Long Training Field (LTF) at WiFi transmitters (Tx) to scramble signals before transmission [11]–[14]; also, there are methods to reconstruct WiFi Rx as full-duplex devices to simultaneously receive signals for legitimate sensing and retransmit them after signal scrambling [15], [16]. Although these methods can effectively distort private information carried by WiFi signals, they require modifying standard WiFi Tx or Rx, which are incompatible with commercial off-the-shelf (COTS) WiFi devices and WiFi standards.

Recent efforts try to scramble WiFi signals by deploying intermediate devices (e.g., RIS) in the sensing environment. These devices can manipulate the wireless channel and sufficiently distort private information [17], [18] to thwart eavesdropping. Also, they operate independently of WiFi devices and standards. However, this independence and constrained resources of these devices result in an indiscriminate scrambling, which disables both eavesdropping and legitimate sensing.

In this paper, we present RISEC¹, a novel hardware–software

¹RISEC is the abbreviation of **R**econfigurable **I**ntelligent **S**urface based **s**ecurity mechanism.

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co-design to simultaneously thwart eavesdropping and preserve legitimate sensing. The key insight behind RISEC is exploiting a RIS with inherent channel programming capability to apply random phase modulation to WiFi signals. This random modulation acts as a dynamic physical mask that scrambles signals and distorts private information. Legitimate users, with essential knowledge, can uncover this mask and accurately recover sensing information. However, realizing this idea is non-trivial and presents several challenges.

1) *Residual Information Leakage*: Only the signals reflected by the RIS are modulated and scrambled, while the private information may still leak from WiFi signals that propagate along other paths and leaving a hole for eavesdroppers.

2) *Blind Dynamic Mask*: As the RIS operates independently and lacks direct communication capabilities, the dynamic mask remains invisible to legitimate users. Therefore, it is difficult for legitimate users to know how each WiFi packet is scrambled and accurately recover the sensing information.

3) *Unexpected Signal Superposition*: Even when legitimate users obtain the knowledge of the dynamic mask, accurately recovering sensing information from the superposed RIS-initiated and Tx-initiated signals remains challenging.

To address these challenges, we integrate the following core designs in RISEC. First, we delicately devise the dynamic mask in both time and spatial domains to ensure its temporal pseudorandomness and wide spatial coverage for sufficient signal scrambling in the sensing area. Second, we encode essential parameters in the mask to help legitimate users uncover the mask and know how each WiFi packet is scrambled. Third, we develop an algorithm for legitimate users to accurately extract sensing results from the superposition of RIS-initiated and Tx-initiated signals once they uncover the dynamic mask.

Our contributions can be summarized as follows:

- We propose RISEC, a novel protection framework for WiFi human sensing that transforms a RIS into a physical mask to thwart eavesdropping and preserve legitimate sensing.
- We devise a hardware-software co-design to tackle the core challenges in operating a RIS-based secure WiFi human sensing system, including residual information leakage, blind dynamic mask, and unexpected signal superposition.
- We implement RISEC and evaluate its performance under various settings. The experimental results show that RISEC significantly reduces an eavesdropper's sensing accuracy to around random-guess levels in password inference (from 94.82% to 0.25%) and activity recognition (from 92.38% to 19.36%), respectively. Moreover, RISEC can also well preserve the legitimate sensing and communication services.

II. RELATED WORK

1) *Secure WiFi Sensing*: The inherently open nature of WiFi signals could be maliciously exploited by eavesdroppers to infer private information. To thwart such eavesdropping, many secure WiFi sensing techniques have been proposed [11]–[19]. Some works focus on delicately scrambling WiFi signals with modified WiFi devices. For instance, WiShield [12] and MIMOcrypt [11] exploit the MIMO capabilities of WiFi devices by assigning unique weights to

each Tx antenna for scrambling transmitted signals. Then, legitimate users can use a dedicated deep learning model to recover sensing information. WiCloak [13] directly modifies the LTF of each WiFi packet to protect location information. Besides, PhyCloak [15] and Aegis [16] respectively employ a full-duplex device to simultaneously receive signals for legitimate sensing and subsequently re-transmit them after scrambling signal characteristics. Although these approaches achieve robust protection, their requirement for reconstructing WiFi hardware compromises device integrity and renders them incompatible with existing WiFi standards and COTS WiFi devices. To avoid modifying WiFi hardware, IRShield [17] deploys a RIS and randomizes its configuration to program signal characteristics, and thus distorts the private information. However, its indiscriminate strategy disables both eavesdropping and legitimate sensing. RF-Protect [20] also proposes a RIS-based method to defend against active radar eavesdropping, but it is not compatible with WiFi signals.

Different from these prior efforts, RISEC protects WiFi human sensing against eavesdropping without requiring any modification to WiFi devices or hindering legitimate sensing.

2) *RIS-assisted WiFi Sensing*: RISs are 2D artificial structures made up of regularly arranged sub-wavelength unit cells [21]–[29]. Their ability to flexibly manipulate wireless signals has been recently integrated with WiFi sensing technologies to enhance both sensing performance and potential applications. For instance, RFlens [22] leverages an RIS with beamforming capabilities to achieve non-line-of-sight human sensing; MetaSense [25] exploits the RIS's diverse radiation patterns for fine-grained motion recognition; MASS [26] designs a dedicated RIS to simulate human motion effects on WiFi signals for sensing datasets augmentation; and Zhou *et al.* [27] uses a RIS with the beam-scanning capability to perform multi-person respiration detection.

Different from the existing works, RISEC employs a RIS for protecting WiFi human sensing, one of the key challenges in existing WiFi sensing technologies and applications.

III. THREAT MODEL

Without loss of generality, we consider a practical scenario in RISEC: an eavesdropper (Eve) passively captures WiFi signals from a WiFi access point (AP) to infer private information (*e.g.*, passwords and daily activities) about a nearby victim (Bob). Eve appears as an ordinary user equipped with a commercial WiFi device and maintains stealthiness during eavesdropping. An eavesdropping is deemed successful once Eve reconstructs the targeted private information. To thwart such eavesdropping, Bob deploys a RIS that applies pseudorandom phase modulations to scramble WiFi signals. Bob can accurately recover the sensing information by employing a dedicated algorithm. Note that both the AP and the RIS are always considered as trusted parties and under Bob's control. We endow Eve with the following capabilities:

1) *Eavesdropping Position*: Different from the existing approaches that require a strict spatial separation between Bob and Eve [16], [17], we don't impose such a requirement. That is, Eve and Bob can coexist in the same room. Nevertheless,

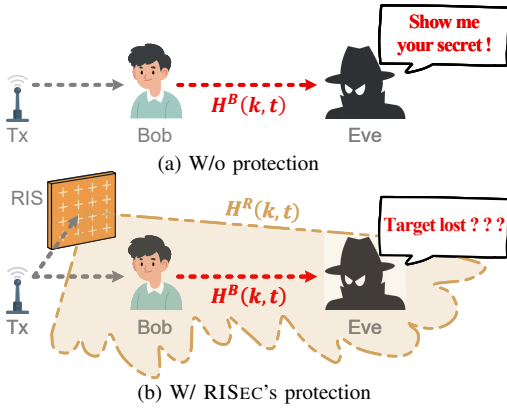


Fig. 2. Brief illustration of RISEC's signal model.

Eve should avoid staying near the AP and RIS for long periods to maintain stealthiness.

2) *Eavesdropping Device*: We assume that Eve carries a portable COTS WiFi Rx (e.g., smartphone or laptop) equipped with a standard network interface card (NIC). Consistent with the configuration of contemporary WiFi NICs [30], we assume that Eve's Rx has three omnidirectional antennas.

3) *Eavesdropping Duration*: Eve can perform continuous and long-term eavesdropping, where the duration is limited only by operational constraints of Eve's eavesdropping device.

4) *Sensing Models*: To follow the open-design principle (i.e., the security of a mechanism should not rely on the secrecy of its design or implementation), we assume that Eve can exploit the same sensing models as Bob, since many existing sensing models are publicly available online. In addition, we also assume that Eve is aware of the RIS deployed by Bob for protection and the basic design of RISEC.

IV. OVERVIEW

A. Design Goals

RISEC aims to achieve the following three goals: First, thwarting Eve's attempt to infer Bob's private information. Second, guaranteeing that Bob can accurately recover the sensing information. Third, operating seamlessly with Bob's WiFi devices and avoiding disrupting WiFi communication.

B. Signal Model

To eavesdrop on Bob's private information, Eve first uses a WiFi Rx to capture WiFi packets and estimate the CSI. Specifically, Eve estimates CSI by calculating $h(k, t) = y(k, t)/x(k, t)$, where $x(k, t)$ is a predefined symbol transmitted on the k -th subcarrier at time t , $y(k, t)$ is the received signal and $h(k, t)$ denotes the resulting CSI estimation.

In the unprotected scenario (illustrated in Fig. 2(a)), the channel response (i.e., propagation matrix) in Bob's environment can be expressed as

$$\mathbf{H}^B(k, t) = \underbrace{\sum_{l^D \in \mathcal{L}_t^D} \mathbf{G}_{l^D}(k, t)}_{\text{Tx-initiated dynamic paths}} + \underbrace{\sum_{l^S \in \mathcal{L}_t^S} \mathbf{G}_{l^S}(k, t)}_{\text{Tx-initiated static paths}} + \underbrace{\mathbf{N}(k, t)}_{\text{noise}},$$

(1)

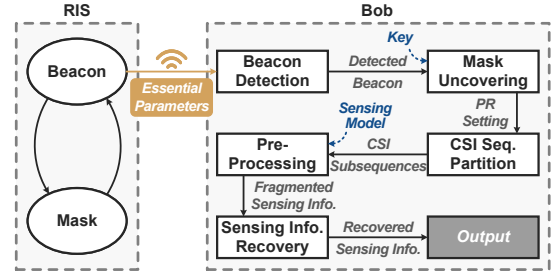


Fig. 3. The high-level workflow of the RIS and Bob.

where $\mathcal{L}_t^D$ and $\mathcal{L}_t^S$ are the sets of Tx-initiated dynamic and static propagation paths, respectively; $\mathbf{G}_{l^D}(k, t)$ and $\mathbf{G}_{l^S}(k, t)$ denote the propagation matrices of the l^D -th dynamic path and l^S -th static path, respectively; $\mathbf{N}(k, t)$ denotes noise. Here, dynamic paths refer to signal propagation paths with reflections from moving objects (e.g., a walking person), while static paths comprise the direct path between the Tx and Rx and reflections from stationary objects (e.g., walls and floors).

As $\mathbf{G}_{l^D}(k, t)$ is mainly influenced by moving objects, it carries most of Bob's private information and thus constitutes Eve's main inference source. In this unprotected scenario, Eve's CSI estimation $\hat{\mathbf{H}}(k, t) = \mathbf{H}^B(k, t)$, which means that Eve can directly access Bob's private information.

To thwart such eavesdropping, RISEC employs a RIS to apply dynamic phase modulation to the WiFi signals. The RIS functions as an unpredictable mask to scramble the wireless channel and thus distort $\mathbf{H}^B(k, t)$ (shown in Fig. 2(b)). Mathematically, by delicately configuring the phase coefficient $r_m(t)$ for each of the M RIS unit cells, RISEC introduces an additional channel component $\mathbf{H}^R(k, t)$, formulated as

$$\mathbf{H}^R(k, t) = \underbrace{\sum_{m=1}^M r_m(t) \mathbf{G}'_m(k, t)}_{\text{RIS-initiated propagation paths}}, \quad (2)$$

where $\mathbf{G}'_m(k, t)$ is the propagation matrix of signals along the m -th RIS unit cell. Now Eve's CSI estimation becomes

$$\hat{\mathbf{H}}'(k, t) = \mathcal{F}[\mathbf{H}^B(k, t), \mathbf{H}^R(k, t)], \quad (3)$$

where $\mathcal{F}[\cdot]$ denotes the complex superposition of Tx-initiated and RIS-initiated signals. Since $\mathbf{H}^R(k, t)$ is determined by the dynamic and secret phase modulation of the RIS, $\mathbf{H}^B(k, t)$ is significantly distorted, and Eve can not extract $\mathbf{H}^B(k, t)$ from $\hat{\mathbf{H}}'(k, t)$. Thus, Bob's private information can be protected.

C. Workflow

Fig. 3 shows the workflow of RISEC, including a RIS for signal scrambling and the legitimate user, Bob.

1) *RIS*: The RIS cycles between modulating the dynamic mask to scramble signals and modulating beacons to help legitimate sensing. To sufficiently scramble WiFi signals, we propose a two-fold design of the mask (Sec. V-A): on one hand, the dynamic mask is designed to change pseudorandomly in time domain to guarantee sufficient variability and unpredictability; on the other hand, the dynamic mask is

optimized in spatial domain to cover a wide angular range for thwarting possible eavesdropping from different directions.

Besides, to preserve the legitimate sensing service for Bob, the RIS intermittently modulates WiFi signals with a beacon. The beacon is also the phase modulation that is optimized in time and spatial domains. The beacon encodes the necessary information (*i.e.*, the pseudorandom setting of the mask) for recovering the sensing information. The configuration of the beacon is pre-shared with Bob through a secure channel and remains inaccessible to Eve. Bob can detect and decode the beacon by executing cross-correlation between the received signals and the beacon template (Sec. V-B1).

Also, we integrate an envelope detector onto the RIS controller to manage the per-packet phase modulation, so as to avoid degrading the WiFi communication quality (Sec. VI).

2) *Bob*: Upon detecting the beacon, Bob employs the private key to uncover the dynamic mask's pseudorandom setting and then segments the captured CSI into several sub-sequences, each corresponding to one of the static mask patterns. This segmentation confines the RIS to a static multipath component within each sub-sequence, thereby allowing Bob to decouple the superposed RIS-initiated and Tx-initiated signals. To recover complete sensing information, we develop a dedicated algorithm to fuse the fragmented information in each sub-sequence for accurate legitimate sensing (Sec. V-C).

Signal Manipulation Process. Based on the above discussion, we briefly describe the signal manipulation processes at RIS, Bob, and Eve's ends.

The deployed RIS, to defend against eavesdropping, configures the phase shifts of its unit cells to introduce random phase shifts on incident WiFi signals, thereby scrambling the signals and the corresponding sensing information they carry.

Bob, as a legitimate user, first detects the periodic beacons through cross-correlation to infer the pseudorandom sequence and the dynamic mask configuration. Then, Bob applies the proposed dedicated algorithm to recover the complete sensing information. For password inference, Bob extracts the temporal variations induced by finger motions and key-strokes from the recovered results, and then feeds them into a keystroke inference model (*e.g.*, WINK [10]). For activity recognition, Bob extracts the Doppler frequency spectrum associated with the activity from the recovered results and then uses it as the input to an activity recognition model (*e.g.*, WiDar3.0 [31]).

As for Eve, due to the lack of knowledge about how the RIS scrambles WiFi signals, Eve directly feeds the CSI extracted from received signals into existing sensing algorithms (*e.g.*, WINK or WiDar3.0). As the sensing information obtained by Eve is severely distorted, the sensing performance is poor.

V. RISEC DESIGN

In this section, we elaborate on the core design components of RISEC, including the design of the dynamic mask and beacon, and the algorithm for sensing information recovery.

A. Dynamic Mask Design

The degree of protection depends on the scrambled signals' unpredictability and coverage. For sufficient scrambling, we

optimize the dynamic mask in time and spatial domains to maximize its temporal randomness and angular coverages.

1) *Pseudorandomization for Temporal Variability*: To introduce sufficient signal scrambling and robustly protect private information, it is not enough to employ a static mask. If so, the RIS will act as a static reflector, and the dynamic signal paths $\mathbf{G}_{ID}(k, t)$ (refer to Eq. (1)) remain dominated by human activity, which may cause the leakage of private information. To address this problem, we introduce temporal variability by pseudorandomly switching between different phase modulations (*i.e.*, masks), so as to make $\mathbf{H}^R(k, t)$ and $\hat{\mathbf{H}}'(k, t)$ highly unpredictable and significantly scrambled.

Specifically, RISEC employs the Salsa20 stream cipher [32] to produce a binary pseudorandom sequence. Each bit of this sequence selects one of two predefined masks, Mask-*A* or Mask-*B*, thereby creating a dynamically changing and unpredictable RIS mask. Each WiFi packet will be scrambled by the mask corresponding to its sequence bit (*e.g.*, Mask-*A* for bit "0" and Mask-*B* for bit "1", or vice versa). Thanks to its lightweight design, Salsa20 can be efficiently implemented on the RIS controller without imposing excessive computational or power overhead². At its core, Salsa20 maintains a 512-bit internal state, which comprises a 256-bit secret key $[K_0 K_1 K_2 K_3 K_4 K_5 K_6 K_7]$ (retained by Bob), a 64-bit nonce $[N_0 N_1]$, a 64-bit block counter $[B_0 B_1]$, and a 128-bit constant $[C_0 C_1 C_2 C_3]$. These parameters are arranged into the following 4×4 state matrix:

$$\mathcal{S} = \begin{bmatrix} C_0 & K_0 & K_1 & K_2 \\ K_3 & K_4 & K_5 & K_6 \\ K_7 & N_0 & N_1 & B_0 \\ B_1 & C_1 & C_2 & C_3 \end{bmatrix}, \quad (4)$$

where each element in $\mathcal{S}$ is a 32-bit word. To generate pseudorandom sequences, Salsa20 applies the quarter-round transformation to four 32-bit words (a, b, c, d) : $b \oplus = (a + d) \lll 7$, $c \oplus = (b + a) \lll 9$, $d \oplus = (c + b) \lll 13$, $a \oplus = (d + c) \lll 18$, where $\lll n$ denotes a left rotation by n bits.

By applying the quarter-round transformation iteratively to each of the four columns and the wrap-around diagonals of the state matrix $\mathcal{S}$, we can obtain a thoroughly diffused state $\tilde{\mathcal{S}}$. $\tilde{\mathcal{S}}$ is then added word-by-word to the original state $\mathcal{S}$ to produce a 512-bit pseudorandom sequence. To continue the generation, the block counter $[B_0 B_1]$ will be incremented by one, and the algorithm will repeat the above process. As the maximum value of the counter is $2^{64} - 1$, this means that the length of a non-repeating pseudorandom sequence is $2^{64} \times 512$. At a WiFi packet rate of 500 Hz, for example, this sequence remains unique for approximately 5.99×10^{11} years, which can cover the whole operational lifetime of the RIS.

Fig. 4(a) illustrates the temporal property of a segment of the pseudorandom sequence. In this segment, each binary output is either mapped to Mask-*A* or Mask-*B*, which introduces significant unpredictability to the mask for sufficiently scrambling WiFi signals and distorting corresponding private information.

²Note that RISEC is a general framework, so it is not limited to any specific number of masks or the choice of pseudorandom sequence generation algorithm. Users can adapt these configurations by themselves based on their unique security requirements and hardware constraints.

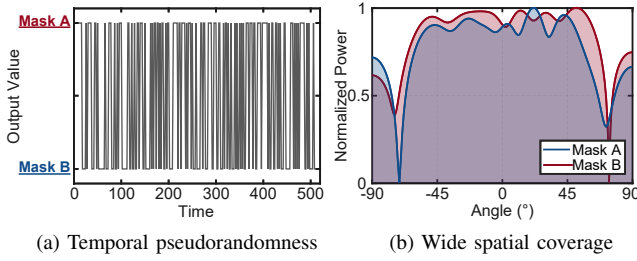


Fig. 4. The properties of the dynamic mask in time and spatial domains.

2) *Optimization of Spatial Coverage*: To ensure the widest possible coverage of the masks, we adopt Genetic Algorithm to optimize the phase shift distribution across the RIS's unit cells (i.e., $\mathbf{r} = \{r_m\}_{m=1}^M$) corresponding to the Mask-A and Mask-B. Our design balances the following two objectives:

On one hand, we seek the largest angle range Ω_{in} to ensure the widest mask coverage, over which the energy of scrambled signals remains as uniform as possible, which is subject to

$$C_1(\mathbf{r}) = \frac{\max_{\theta \in \Omega_{in}(\mathbf{r})} P(\theta; \mathbf{r})}{\min_{\theta \in \Omega_{in}(\mathbf{r})} P(\theta; \mathbf{r})} < \epsilon_1, \quad (5)$$

where $P(\theta; \mathbf{r})$ denotes the energy distribution at angle θ under the phase shift distribution $\mathbf{r}$, and ϵ_1 is a prescribed threshold.

On the other hand, the marginal effect of RISs [6], [22] may lead to an obvious energy drop in very large angles, where we denote these angles as $\Omega_{out} = \Theta - \Omega_{in}$, $\Theta \in [-90^\circ, 90^\circ]$. To minimize the energy difference between Ω_{out} and Ω_{in} , we enforce the following constraint:

$$C_2(\mathbf{r}) = \frac{\frac{1}{|\Omega_{out}(\mathbf{r})|} \int_{\Omega_{out}(\mathbf{r})} P(\theta; \mathbf{r}) d\theta}{\frac{1}{|\Omega_{in}(\mathbf{r})|} \int_{\Omega_{in}(\mathbf{r})} P(\theta; \mathbf{r}) d\theta} < \epsilon_2, \quad (6)$$

where the threshold ϵ_2 bounds allowable energy difference³.

By combining the above two constraints, we can formulate the overall optimization problem as follows

$$\mathbf{r}^* = \arg \max_{\mathbf{r}} |\Omega_{in}| \quad \text{s.t.} \quad \begin{cases} C_1(\mathbf{r}) < \epsilon_1 \\ C_2(\mathbf{r}) < \epsilon_2 \end{cases}. \quad (7)$$

By solving the above optimization equations, we can obtain a set of phase shift distributions that satisfy the dual constraints. For example, Fig. 4(b) shows the spatial domain property of two candidate results for Mask-A and Mask-B. Both of the candidate results exhibit approximately uniform coverage over a wide angular span of 120° and an adequate energy difference between Ω_{out} and Ω_{in} . Even though an energy drop is observed over a narrow angular segment, it can be effectively compensated by multipath propagation to maintain signal scrambling throughout the environment. Consequently, Eve can't accurately extract private information from the received signals, even along such marginal directions.

³We set $\epsilon_2 = 0.35$ in RISEC, which can be adaptively adjusted according to Bob's requirements. Note that the RIS used in RISEC controls the phase and angular distribution of the reflected signals, while introducing negligible change to the reflected signal power. Therefore, under the constraint in Eq. (6), the optimized masks would ensure sufficient energy within the covered region and will not leave Ω_{in} with arbitrarily small practical gain.

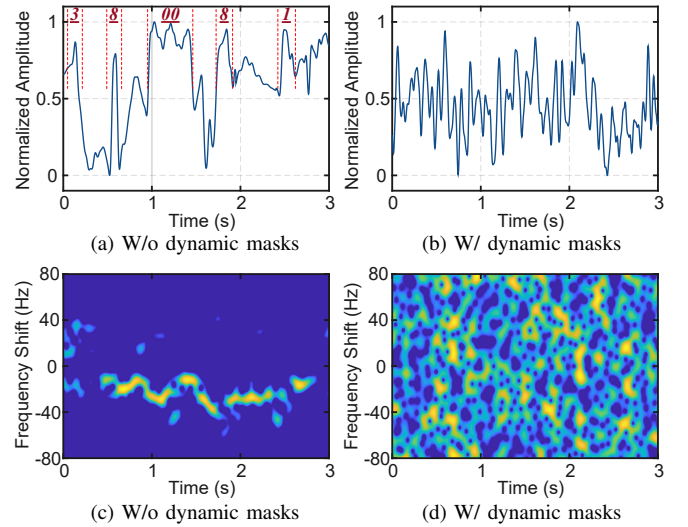


Fig. 5. Sensing results without and with dynamic masks. (a)-(b) Time series of the PIN password "380081"; (c)-(d) DFS of the activity "walking away".

Proof-of-concept Validation. We set up a group of proof-of-concept experiments to validate the effectiveness of the dynamic mask on protecting sensing information:

- **Password inference**: In this group of experiments, a volunteer is asked to use a smartphone to enter the password "380081". The Rx extracts WiFi CSI to analyze the keystroke pattern. As Fig. 5(a) shows, when the RIS is off (i.e., w/o dynamic masks), the keystroke pattern is visible, and Eve may recover the password by using existing techniques [10]. By contrast, as Fig. 5(b) shows, when the dynamic mask is applied to scramble WiFi signals, the features of keystrokes in signals are thoroughly distorted.
- **Activity recognition**: In this group of experiments, a volunteer is asked to walk away from the WiFi Rx. The Rx extracts DFS by using the existing method [33] to analyze the volunteer's motion. As Fig. 5(c) shows, when the RIS is off, the walk-away motion produces clear DFS signatures. By contrast, as Fig. 5(d) shows, when the dynamic mask is applied to scramble WiFi signals, the DFS is deeply distorted and conveys no meaningful information.

Take-away: The above results show that *the RIS, when configured as a dynamic mask, can scramble WiFi signals and effectively protect sensing information against eavesdropping.*

B. Beacon Generation and Detection for Legitimate Sensing

1) *Beacon Generation*: Although dynamic masks effectively thwart eavesdropping by unpredictably scrambling WiFi signals, they also prevent Bob from extracting any sensing information. What's worse, the RIS lacks direct communication capabilities to transmit per-packet mask information to Bob. To overcome these challenges, we leverage the observation that once Bob acquires the 64-bit counter value $[B_0 \ B_1]$, he/she can deduce the output of the pseudorandom sequence by using the secret key and static parameters in the state matrix S .

Leveraging this insight, we introduce a lightweight beacon scheme: the RIS intermittently suspends the output of the pseudorandom sequence and the modulation of the dynamic mask, while turning to modulate the information about the

latest counter value, namely the *beacon*. Specifically, we represent the 64-bit counter $[B_0 \ B_1]$ as $[b_0, b_1, b_2, \dots, b_{64}]$, which we partition into sixteen 4-bit segments. Each segment, ranging from "0000" to "1111", is encoded by one of 16 pre-assigned 128-bit beacon codes. Note that these beacon codes, combined with a 256-bit preamble, are carefully selected binary sequences and securely retained by Bob.

The RIS cycles between modulating the dynamic mask and the beacon. In a beacon round, the change of the phase pattern across RIS unit cells is guided by the following 3 components:

$$\underbrace{\text{preamble}}_{\substack{256\text{-bit} \\ \text{signaling the start}}} \parallel \underbrace{\text{beacon codes}}_{\substack{16 \times 128\text{-bit} \\ \text{encoding the counter value}}} \parallel \underbrace{\text{PR sequence}}_{\substack{q \times 512\text{-bit} \\ \text{modulating the dynamic mask}}}, \quad (8)$$

where $q \in [0, 31]$ is the decimal value of the final 5 bits of the pseudorandom sequence in the preceding beacon round (chosen randomly for the initial round). The variability of q aims to break the fixed periodicity and length of each beacon round (defined in Eq. (8)) for security enhancement.

To modulate the "0" bits and "1" bits in the preamble and beacon codes, we apply a phase pattern across the RIS unit cells that also meets the constraints of spatial coverage formulated in Eq. (7). We denote this phase pattern as Mask- C : it encodes a "0" bit directly, and encodes a "1" bit by inverting every unit cell by π rad (namely Mask- $\tilde{C}$). This inversion doesn't alter coverage and can be detected by Bob via cross-correlation. Due to the temporal variability of the preamble and beacon codes, as well as the wide spatial coverage of Mask- C , the private information is still significantly distorted during the process of beacon modulation.

2) *Beacon Detection*: To detect the beacon, Bob first computes signal differencing between consecutive received signals for denoising. Then, Bob performs cross-correlation between the processed signal with the known template (*i.e.*, binary sequence of the preamble varying between Mask- C and Mask- $\tilde{C}$). Based on the correlation result, Bob can detect and decode each 128-bit beacon code to recover the full 64-bit counter. The length and the corresponding binary sequence of the preamble and beacon codes can be adaptively configured by Bob. Once obtaining the value of the counter, Bob can deduce all future modulations of the RIS (including the binary sequences corresponding to Mask- A and Mask- B in the dynamic mask modulation, as well as the binary sequences corresponding to the preamble and beacon codes). In other words, as long as Bob remains within the sensing area, there is no need to repeatedly detect the beacon. After that, Bob can proceed to extract sensing results from the superposed RIS-initiated and Tx-initiated signals.

C. Sensing Information Recovery

Based on the channel models formulated in Eq. (1)-(3), an intuitive idea for recovering sensing information is to directly subtract the RIS's channel component $\mathbf{H}^R(k, t)$ from the received superposed signals $\hat{\mathbf{H}}'(k, t)$. However, this is extremely challenging: the RIS-initiated signals are deeply coupled with the Tx-initiated signals, making it infeasible to isolate the RIS's impact.

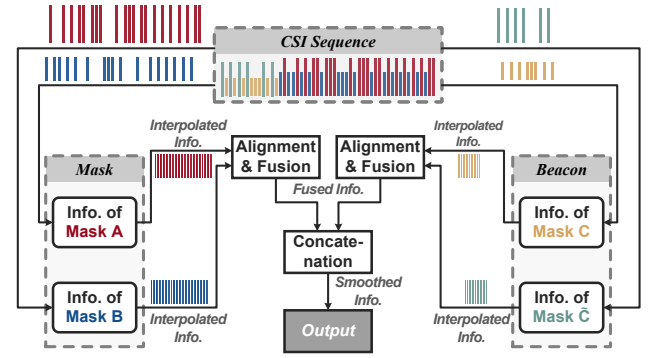


Fig. 6. Process of SIR algorithm.

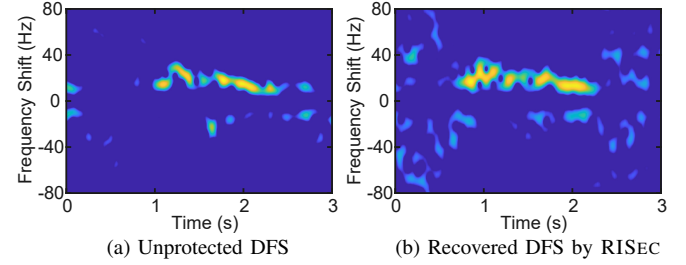


Fig. 7. The recoverability of RISEC for legitimate sensing.

Instead, recall that the RIS in RISEC cycles among four different modulation patterns, which we denote as $P \in \{A, B, C, \tilde{C}\}$. Our key observation is that when the collected CSI samples are partitioned according to these four patterns, the RIS acts as a static reflector within each subset. Specifically, for time indices $t \in \mathcal{T}^P$, the RIS phase coefficients $r_m(t)$ in Eq. (2) remain unchanged. Therefore, the RIS-initiated propagation paths become time-invariant in each subset and can be regarded as part of static components. The channel variation in each subset is then dominated by dynamic paths $\mathbf{G}_{ID}(k, t)$, which capture the user's motion and are exactly the sensing information we wish to extract.

Building on this observation, we can reconstruct the entire sensing information by extracting the fragmented information from all four subsets and then appropriately fusing them. However, it is important to note that we can not directly fuse the raw CSI sequences across subsets because they are captured under different static channel responses (due to the varying r_m^P). In contrast, fusing the extracted sensing information can mitigate the impact of these static channel differences. Specifically, the proposed sensing information recovery (SIR) algorithm (illustrated in Fig. 6) proceeds as follows:

1) *CSI Partition and Sensing Information Extraction*: Taking a beacon round (formulated in Eq. (8)) as an example, Bob first obtains CSI estimations $\hat{\mathbf{H}}'(k, t)$ from the received signals, then groups the CSI sequence into four CSI-subsets based on different modulation pattern: $\mathcal{H}^P = \{\hat{\mathbf{H}}'(k, t) \mid t \in \mathcal{T}^P\}$. Here, $\mathcal{T}^P$ is obtained through packet-level temporal alignment after beacon detection. Specifically, Bob first detects the beacon by cross-correlating the received CSI with the beacon codes. Once the correlation peak is detected and the counter is decoded, Bob can determine the start time of the beacon round and further infer the modulation pattern of each subsequent packet. As the channel dynamics within each

subset $\mathcal{H}^P$ are dominated by user movement $\mathbf{G}_{l^D}(k, t)$, Bob can apply sensing algorithms to extract the corresponding sensing information fragment (e.g., DFS) from each subset. Partitioning errors mainly occur when the beacon is not correctly detected or decoded. In this case, the recovered sensing information remains scrambled and unreliable. To avoid such errors, Bob can detect multiple beacon rounds or start SIR when the correlation peak is sufficiently clear.

2) *Interpolation and Fusion of Paired Sensing Information:* To reconstruct continuous sensing information, we interpolate and align the fragments from Mask-A and Mask-B. Specifically, we first employ the Short-time Fourier transform (STFT) to interpolate $\mathcal{I}^A$ and $\mathcal{I}^B$ onto their combined time domain $\mathcal{T}^{AB} = \mathcal{T}^A \cup \mathcal{T}^B$, and then fuse the two interpolated results using the Discrete Wavelet Transform (DWT) to produce an intermediate reconstruction $\hat{\mathcal{I}}^{AB}$. This fusion step is essential: although interpolation temporally aligns the fragments, it does not introduce new information. DWT-based fusion ensures that we fully exploit the complementary information spanning across subsets. A parallel interpolation and fusion process is applied to fragments $\mathcal{I}^C$ and $\mathcal{I}^{\hat{C}}$ to generate $\hat{\mathcal{I}}^{CC}$. In our implementation, STFT uses a Gaussian window of 95 samples, an overlap of 90 samples, and a 2048-point FFT. For DWT, we use Daubechies-4 wavelet with a three-level decomposition.

3) *Concatenation and Smoothing:* Finally, we concatenate $\hat{\mathcal{I}}^{AB}$ and $\hat{\mathcal{I}}^{CC}$ to recover the complete sensing information, formulated as $\hat{\mathcal{I}} = \hat{\mathcal{I}}^{AB} \oplus \hat{\mathcal{I}}^{CC}$. To remove any discontinuities or high-frequency noise introduced during the interpolation and fusion stages, we apply a global smoothing and denoising to $\hat{\mathcal{I}}$. Specifically, we apply a moving-average filter with a window length of 5 STFT frames. This operation smooths temporal artifacts caused by interpolation and concatenation, and suppresses high-frequency noise that is inconsistent with continuous DFS features. The SIR algorithm then outputs the final sensing information and delivers it to Bob for follow-up sensing tasks. Note that the SIR algorithm can be applied to different sensing applications, with only minor parameter adjustments when necessary. For example, respiration monitoring can use a larger STFT window to improve frequency resolution for subtle low-frequency breathing patterns, while trajectory tracking can adopt Kalman filter for smoothing to suppress position jitter.

Proof-of-concept Validation. We conduct a quick validation to evaluate the effectiveness of the proposed algorithm. A volunteer is asked to walk towards the WiFi Rx. The Rx extracts the DFS to analyze the volunteer's motion. As shown in Fig. 7(b), the DFS recovered by RISEC faithfully preserves the activity's characteristics (i.e., standing up). Although slight noise remains, it does not affect the overall trend and pattern of the DFS, which can still be accurately classified by the pre-trained sensing model.

D. Security Analysis

In this section, we present a brief theoretical analysis of the security of RISEC. As discussed before, the RIS-based dynamic mask is driven by a pseudo-random sequence, which injects high-entropy and unpredictable scrambling into

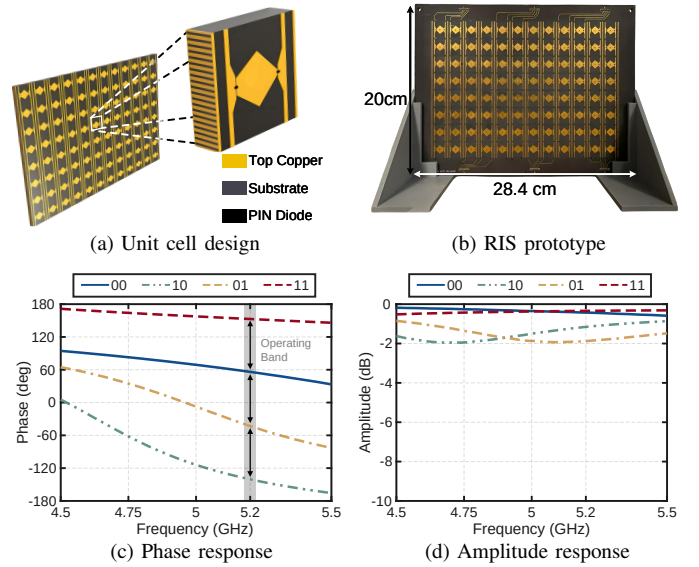


Fig. 8. The RIS designed in RISEC.

signals. Due to the packet-level pseudorandom scrambling, the variation introduced by the RIS, $\text{Var}[\mathbf{H}^R(k, t)]$, is significantly larger than the variation caused by human motion, $\text{Var}[\sum_{l^D \in \mathcal{L}_t^D} \mathbf{G}_{l^D}(k, t)]$. As a result, the dynamic component carrying sensing information is submerged in the stronger randomized scrambling. Consequently, the mutual information between Eve's received signals and Bob's private information approaches zero, that is, $I(\hat{\mathbf{H}}'(k, t); \mathbf{H}^B(k, t)) \approx 0$. This establishes the security foundation that Eve's observation reveals almost no useful information about Bob's private information.

If Eve attempts to remove $\mathbf{H}^R(k, t)$, there will be a severely underdetermined blind source separation problem. At each time snapshot t , Eve must solve for $M+1$ unknowns, including the target channel $\mathbf{H}^B(k, t)$ and the M RIS-initiated cascaded paths $\mathbf{G}'_m(k, t)$ with their time-varying states $r_m(t)$. Since the number of observations is far smaller than the number of unknowns, and all components are strongly coupled, recovering $\mathbf{H}^B(k, t)$ without the RIS configuration sequence is mathematically intractable. Moreover, predicting the mask sequence to cancel $\mathbf{H}^R(k, t)$ is also computationally infeasible. The state transitions are generated by the Salsa20 stream cipher. Without the 256-bit secret key, Eve must rely on exhaustive key search with complexity $O(2^{256})$, which is beyond practical reach.

VI. RIS IMPLEMENTATION

The RIS employed in RISEC is a 2D planar array of periodically arranged sub-wavelength unit cells. Fig. 8(a) depicts the structure of the proposed RIS with a detailed view of an individual unit cell. Each cell comprises a regular hexagonal metal patch whose edges are bridged to DC feed lines via a pair of PIN diodes. By adjusting the bias voltage applied to these feed lines, the on-off states of PIN diodes can be precisely controlled to delicately configure $\mathbf{H}^R(k, t)$ and shift the phase of incoming signals. As shown in Fig. 8(b), the prototype of the RIS comprises a 12×8 array of these unit cells, and is sized 28.4 cm $\times$ 20 cm. In RISEC, we configure

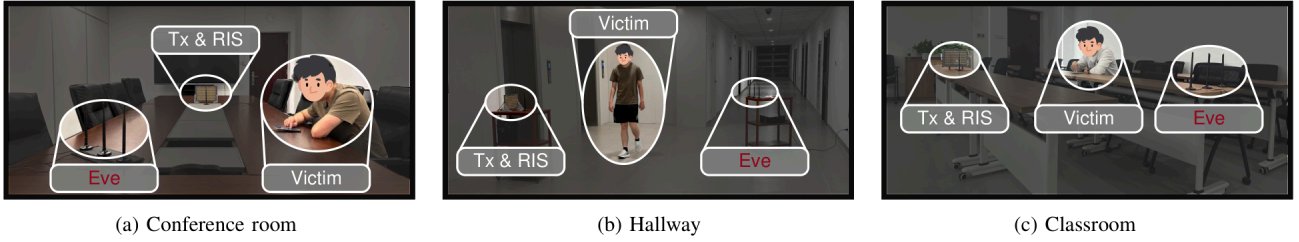


Fig. 9. Experimental scenarios and deployments in RISEC.

the RIS to operate at the 5.2 GHz WiFi band, and phase shifts across unit cells are controlled by an FPGA.

The phase response of the proposed RIS is shown in Fig. 8(c), where four RIS states are taken as examples, with an ideal phase increment of $\pi/2$ between adjacent states (*i.e.*, “00”, “01”, “10”, and “11”). The results show that the phase difference between adjacent states remains close to $\pi/2$, while the relative phase error is bounded within 5%. Such a limited error has negligible impact on the RIS’s modulation. The amplitude response in Fig. 8(d) further shows that the reflection loss introduced by the RIS is less than 2 dB. Moreover, the change of the phase pattern should be carefully controlled to occur in the interval between consecutive WiFi packets to avoid interrupting communication. Thus, we employ an envelope detector [34] to monitor WiFi packets and ensure a precise phase shift application to each WiFi packet.

VII. EVALUATION

A. Experiment Setup

1) *Setting*: Fig. 9 shows the setting of our experiments. We use a pair of WiFi Tx and Rx equipped with COTS WiFi NICs [30] to transmit and receive WiFi signals at channel 40 (*i.e.*, 5.19 GHz-5.21 GHz), respectively. The Tx is equipped with one antenna and the Rx is equipped with three antennas. The WiFi packet rate is 500 Hz. We conduct experiments in a conference room, a hallway and a classroom, respectively. The RIS operates at 5.2 GHz and is placed 15 cm in front of the WiFi Tx. We run PicoScenes [35] to extract CSI data from received WiFi signals, then analyze CSI data and execute sensing algorithms with MATLAB and Python.

2) *Sensing Tasks and Metrics*: We evaluate RISEC’s performance of protecting WiFi human sensing through two practical sensing tasks: password inference and activity recognition. We recruit 8 volunteers to participate in the evaluation and data collection process. Our evaluation is in strict accordance with our institution’s Institutional Review Board (IRB) requirements and has obtained ethics approval and informed consent.

- In password inference, each volunteer is asked to use different smartphones to type a total of 50 predefined 6-digit passwords. Each password is repeated 40 times per volunteer. We employ WINK [10] to evaluate the performance of password inference. Specifically, WINK is a model-driven method by combining the inter-keystroke timing and spatiotemporal features to infer keystrokes. We adopt top- N password inference accuracy as a metric, where an inference is deemed successful if the correct password appears among the top N predicted candidates.

- In activity recognition, each volunteer is asked to perform 6 common activities: walking approach (WA), walking away (WW), pushing and pulling (PP), raising arm (RA), standing up (SU), and sitting down (SD). Each activity is repeated 200 times per volunteer. We employ WiDar3.0 [31] to evaluate the performance of activity recognition. Specifically, WiDar3.0 is a data-driven method by exploring DFS and velocity profiles of gestures to perform gesture recognition. We adopt recognition accuracy as a metric.

3) *Baselines*: We select two methods as the baselines for comparison: WiShield [12] and IRShield [17]. WiShield adjusts MIMO beamformers to scramble WiFi signals and CSI by applying distinct complex values to each transmitting antenna when a WiFi packet is sent, and Bob uses a neural network to recover the CSI. We reproduce WiShield using a USRP X310 equipped with 4 antennas and adjust the WiFi beamformer to scramble WiFi signals. IRShield leverages a RIS and randomizes its configuration to scramble WiFi channel and distort sensing information. We reproduce IRShield using the same RIS as RISEC and configure the RIS elements based on IRShield’s strategy to scramble WiFi signals.

B. Overall Performance

In protecting password inference, as shown in Fig. 10(a), both RISEC and the baseline methods reduce Eve’s inference accuracy to around random-guess levels. Specifically, when employing RISEC for protection, Eve’s Top-1000 accuracy is reduced from 94.82% to 0.254% (*vs.* 0.1% for random guessing) and Top-2000 accuracy from 97.63% to 0.431%. This drastic accuracy decline occurs because both RISEC and the baseline methods scramble the fine-grained CSI features, where the signal variations induced by finger movement are distorted. Thus, Eve can not accurately map received signals back to the corresponding keystrokes.

In protecting activity recognition, Fig. 10(b) demonstrates that RISEC significantly reduces Eve’s recognition accuracy to 19.36%, close to the random-guess level (16.67%). The corresponding confusion matrix is shown in Fig. 10(c). While the other two baseline methods also constrain Eve’s accuracy to approximately 19%, they also impact legitimate sensing. As for IRShield, it does not provide essential information for Bob when applying the indiscriminate scrambling strategy, so Bob’s sensing accuracy also drops to the random guessing level as Eve does. In contrast, due to our dedicated SIR algorithm, Bob can achieve 90.25% accuracy when using RISEC, less than 2% decrease compared to the unprotected scenario (yellow dashed line in Fig. 10(b)).

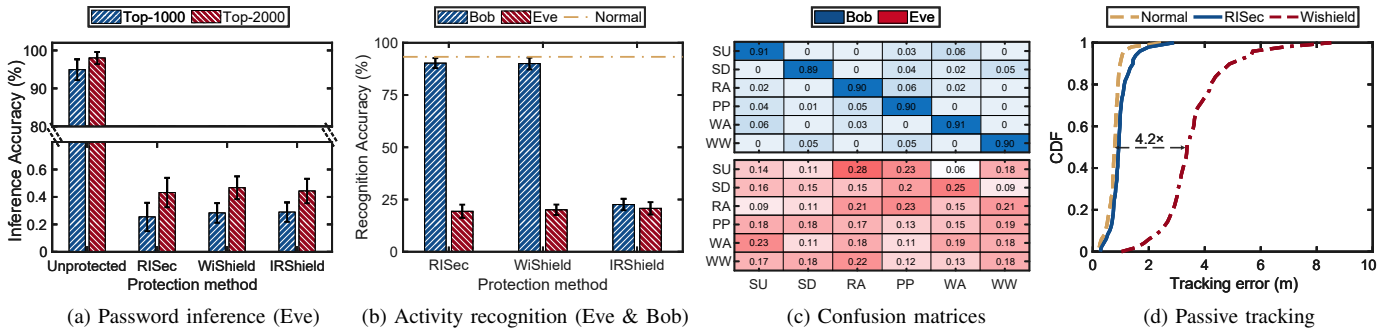


Fig. 10. Overall performance of RISEC.

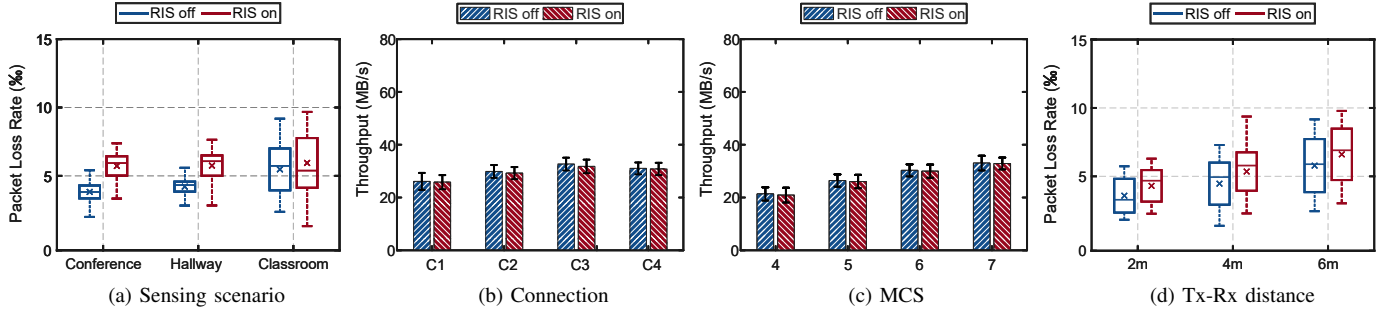


Fig. 11. Impact on WiFi communication..

As for WiShield, although it can help Bob achieve 90.17% accuracy in activity recognition, it may not generalize well to model-driven sensing tasks [12]. As an example shown in Fig. 10(d), in human tracking task, WiShield raises the median tracking error by over $4.2 \times$. In contrast, RISEC yields only a 0.13 m increase in error over the unprotected scenario [33] (*i.e.*, 0.79 m shown by the yellow dashed line). The main reason for this difference is that WiShield depends on a data-driven recovery model whose performance is sensitive to the training data distribution and task type. Although such a method can extract categorical features that are sufficient for activity recognition, flawlessly restoring the exact spatial-temporal signal features remains a challenge.

C. Impact on WiFi Communication

We evaluate the impact of RISEC's operation on WiFi data communication. With the AP's packet rate set to 500 Hz, we record the Rx's packet loss rate (PLR) when the RIS is on and off, respectively. As Fig. 11(a) shows, across different scenarios, the average PLR does not increase drastically after the RIS is on (less than 2%). We also evaluate the impact of RISEC on the throughput of WiFi links. We use different smartphones to establish connections with two nearby APs (AP1 and AP2) around the sensing area and record the variation in throughput over 30 minutes for five links: C1 (Tx-Rx), C2 (iPhone 14 Pro-AP1), C3 (Xiaomi 13 Pro-AP1), C4 (iPhone 14 Pro-AP2), and C5 (Xiaomi 13 Pro-AP2). Results in Fig. 11(b) show that average throughput decreases by no more than 0.8 MB/s after the RIS is on. Further, we evaluate the communication performance under different Modulation and Coding Schemes (MCSs) and Tx-Rx distances. Fig. 11(c) shows that the throughput increases as MCS varies from 4 to 7, while Fig. 11(d) shows that PLR increases as the Tx-Rx distance grows from 2 m to 6 m. In both cases, the

communication performance remains stable when the RIS is switched on or off. These results indicate that RISEC has a negligible impact on WiFi communication.

D. Performance Against Adversarial Attacks

We evaluate the performance of RISEC against potential adversarial attacks launched by Eve.

1) *Clustering Attack*: Eve may analyze the CSI variation between consecutive WiFi packets to infer RIS states. Specifically, Eve may assume that two consecutive CSI samples belong to the same RIS mode if variation is smooth, and to different RIS modes if an abrupt change appears. Thus, Eve can segment the CSI sequence into two states and apply the same strategy as Bob to recover private information.

We conduct experiments to evaluate the feasibility of this attack. Specifically, Eve first collects and pre-processes the CSI sequence. Then, Eve applies principal component analysis and retains the first 10 principal components to construct CSI feature vectors. To analyze CSI temporal continuity, Eve measures the normalized distance between the feature vectors of consecutive packets, and applies K-means clustering with ($K=2$) to divide them into two categories: "same state" and "different states". After that, Eve segments the CSI sequence and applies the same sensing pipeline as Bob.

Fig. 12(a) shows Eve's sensing accuracy under this clustering attack in the activity recognition task. The results show that Eve's sensing accuracy remains close to the random-guess level. The reason is that, even when two consecutive packets are modulated by the same RIS mask, the CSI variation may still be significant due to human motion. Conversely, when the RIS switches to a different mask, the abrupt change may be weakened by destructive superposition. As a result, it is difficult to reliably infer the RIS state by this clustering attack.

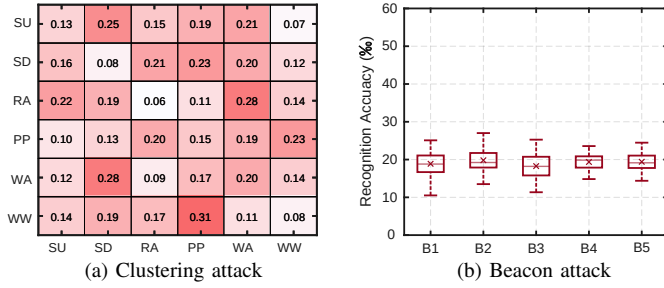


Fig. 12. Adversarial attack.

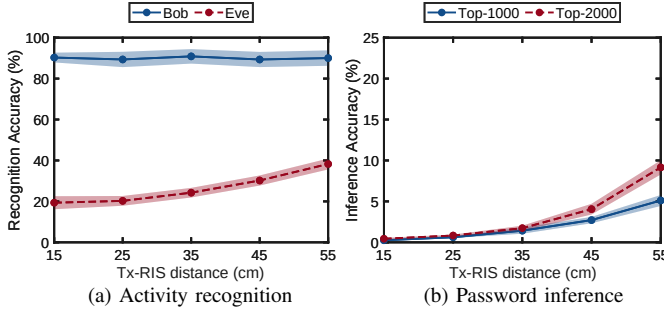


Fig. 14. Impact of different Tx-RIS distances.

2) *Beacon Attack*: Eve may also attempt to infer sensing information from the CSI collected during the beacon interval, since private activities may occur during beacon modulation. However, the beacon codes and beacon length are securely retained by Bob and kept secret from Eve. Thus, Eve can't employ the correlation-based strategy to detect the beacon. A more practical strategy for Eve is to feed the masked CSI into the sensing model and attempt to infer sensing information.

We conduct experiments to evaluate the feasibility of this attack. Specifically, Eve first collects and pre-processes the CSI sequence without knowledge of the beacon codes or beacon length. Then, Eve directly feeds the CSI collected during beacon modulation into the activity recognition model. In this experiment, we assume that Eve knows the start time of the beacon for conservative evaluation, but this information is unavailable to Eve in practice.

Fig. 12(b) shows Eve's sensing accuracy under 5 different beacon sequences in the activity recognition task. The results show that Eve's sensing accuracy remains close to the random-guess level for all beacon sequences. The reason is that the beacon sequence itself also serves as a secret and temporally varying physical mask. Without the knowledge of beacon codes, Eve can't align the received CSI with the RIS states, so the sensing information remains scrambled. As a result, even though the beacon may occupy a relatively long interval, Eve still can not obtain accurate sensing information.

E. Impacting Factors

We evaluate RISEC's robustness and applicability by investigating how different factors influence its performance.

1) *Impact of Different Angles of Eve's Rx*: Eve may attempt to eavesdrop from any possible angle to infer Bob's sensing information. To evaluate the robustness of RISEC, we vary the

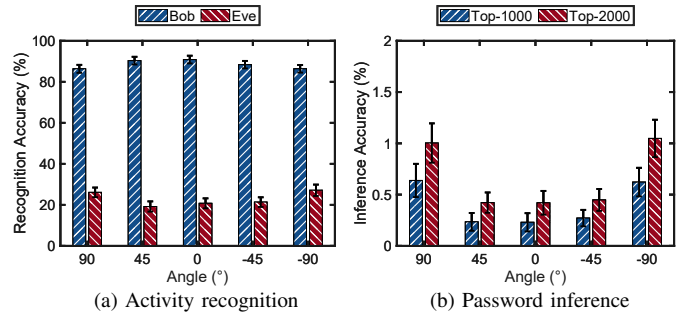


Fig. 13. Impact of different angles of Eve's Rx.

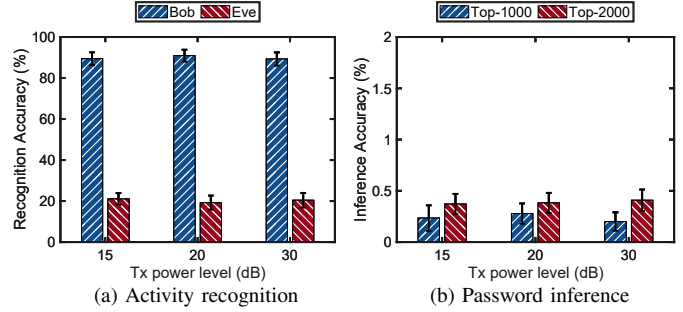


Fig. 15. Impact of different Tx power levels.

Rx's angle relative to the RIS. For a worst-case evaluation, we set the Rx and Bob in the same direction. As Fig. 13 illustrates, Eve's sensing accuracy is reduced to around random-guess levels across most angles, but slightly increases at around $\pm 90^\circ$. This is mainly due to the RIS's reduced reflectivity at extreme edge directions. Even so, Eve's accuracy does not exceed 27.11% in activity recognition, and remains below 1.048% in password inference in such angles. Meanwhile, Bob still achieves around 90% accuracy in activity recognition.

2) *Impact of Different Tx-RIS Distances*: As the RIS operates independently of the WiFi Tx, the Tx-RIS distance determines the proportion of WiFi signals that are scrambled by the RIS, and thus, the degree of protection. We vary the Tx-RIS distance to evaluate its impact on RISEC's performance. As Fig. 14 shows, when the distance increases modestly from 15 cm to 25 cm, Eve's sensing accuracy stays around random-guess levels (below 20.26% in activity recognition and below 0.62% in password inference), indicating that sufficient signal are scrambled to thwart eavesdropping. However, as the Tx-RIS distance grows from 25 cm to 55 cm, we observe a clear increase in Eve's accuracy. The reason for this increase is two-fold: first, a smaller fraction of the WiFi signals is reflected and scrambled by the RIS; second, the energy of the scrambled signals decreases with distance.

3) *Impact of Different Tx Power Levels*: Then, we evaluate the performance of RISEC under several commonly used WiFi power levels (e.g., 15 dBm, 20 dBm, and 30 dBm). As Fig. 15 illustrates, the variation in Tx power level shows negligible influence on RISEC's performance, which means that RISEC can seamlessly work with existing WiFi devices and does not require additional power level calibration.

4) *Impact of Different Number of the RIS's Unit Cells*: We also evaluate RISEC's performance as the number of activated RIS unit cells varies. As Fig. 16 shows, increasing the number

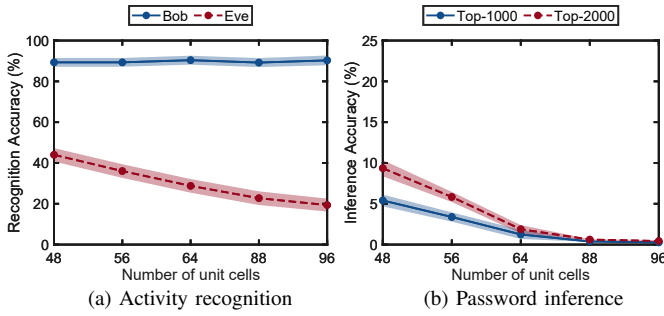


Fig. 16. Impact of different numbers of unit cells.

TABLE I
PROCESSING DELAY ANALYSIS

Device	Operation	Mean Processing Delay
Rx	SIR Algorithm-Step 1	106.32 ms
Rx	SIR Algorithm-Step 2	47.15 ms
Rx	SIR Algorithm-Step 3	29.58 ms
RIS	512-bit PR sequence generation	0.45 μ s
RIS	Beacon generation	0.12 μ s

of activated unit cells leads to a significant drop in Eve's sensing accuracy. Specifically, with only half of the cells activated, Eve's accuracy rises to 43.79% in activity recognition and 9.36% in Top-2000 password inference. However, once over 90% of the unit cells are activated, Eve's performance again falls to around random-guess levels. This trend reflects the fact that more activated unit cells will lead to more significant signal scrambling, and in turn, more robust protection.

5) *Impact of Different Scenarios*: To evaluate the scalability and applicability of RISEC across different sensing scenarios, we conduct experiments in the above-mentioned three scenarios, including a conference room, a hallway, and a classroom. As Fig. 17 illustrates, RISEC delivers robust protection against eavesdropping, and simultaneously, preserves legitimate recoverability in these scenarios. These results underscore RISEC's potential for practical deployment in different indoor settings.

F. Computational Efficiency Analysis

The processing delay of RISEC is reported in Table I. At the Rx side, the three steps of the SIR algorithm take 106.32 ms, 47.15 ms, and 29.58 ms on average, respectively. Step 1 dominates the overall cost because it involves time-consuming sensing-related processing, such as DFS computation. Even so, the overall delay remains at a sub-second level. At the RIS side, the processing overhead is negligible. The mean delay for generating a 512-bit Salsa20 pseudorandom sequence is 0.45 μ s, while that for beacon generation is only 0.12 μ s.

VIII. DISCUSSION

1) *Hardware Cost*: It is reasonable to expect continuous decline of the deployment cost of RIS, considering that this technology is under fast development and will become mature in the coming years. Besides, in our design, the envelope detector costs less than \$10, with minimal additional cost.

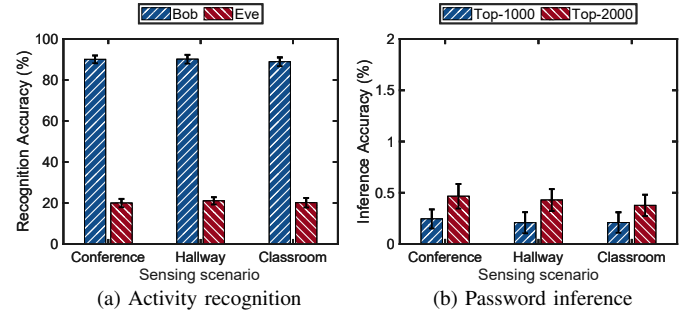


Fig. 17. Impact of different scenarios.

2) *Scalability to Multiple Users*: In a multi-user scenario, one can first remove the dynamic mask by using the SIR algorithm, and then employ different techniques, *e.g.*, blind source separation [36] or data-driven methods [37], [38], to separate multiple users' sensing information at the signal level.

IX. CONCLUSION

We present RISEC, a novel protection framework for WiFi human sensing. By fully exploiting the RIS's channel programmability to apply random phase modulation, RISEC transforms the RIS into a dynamic physical mask that scrambles WiFi signals to protect private information. We propose a delicate hardware-software co-design to reduce an eavesdropper's sensing accuracy to around near random-guess levels, while simultaneously preserving legitimate sensing and communication services. We envision RISEC as a practical security solution to meet secure demands of future applications.

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